

Eloquent lab proposal for CLEF 2025

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Large language models (LLMs) are a foundational technology for both research and information service development, and are used as a tool for solving challenging text analysis problems, not least as evidenced by their use in many of the CLEF labs.

New models with generative capacity are released with various capabilities and intended usage scenarios. The practical application of such models in public arenas, commercial and elsewhere, is challenged by both engineering concerns and to a great extent by quality concerns about predictability and reliability of the output from a generative component. A language model can provide a system with fluently generated output but without guarantees for trustworthiness with respect to relevance, veracity, or consistency. Quality assurance, evaluation, and benchmarking of large language models is performed using tests that focus on details of language understanding and reasoning, but many factors that are crucial for application development and deployment to third parties are not addressed systematically through existing benchmarks.

The ELOQUENT evaluation lab experiments with new evaluation methods for generative language models to meet some of the challenges in the path from laboratory to application. The organisers include commercially active AI developers as well as research groups. The intention of the lab is to explore the following important characteristics of generative language model quality:

- 1) *Trustworthiness*, a many-faceted notion which involves topical relevance and truthfulness, discourse competence, reasoning in language, controllability, and robustness across varied input.
- 2) *Multi-linguality and cultural fit*: the suitability of a language model for some cultural and linguistic area.
- 3) *Self-assessment*: the reliability of a language model to assess the quality of itself or some other language model, using as little human effort as possible.
- 4) *Limits of language models*: the delimitation of world knowledge and generative capacity.

In a departure from CLEF practice and in keeping with the self-assessment point above, we develop tasks which rely less on human-assessed gold standards to expend minimal human effort for the assessment, instead leveraging the intrinsic qualities of generative AI for test development and scoring.

I. LESSONS FROM THE 2024 EDITION OF ELOQUENT

The 2024 edition was first year of the ELOQUENT lab for evaluating generative language model quality. We presented four experimental tasks. To test (a) *topical competence*, we have defined the *Topical Quiz* task for models to self-assess their knowledge of various topics. The focus of the task is

to assess the reliability of such self-testing. To test (b) *model hallucinations*, we defined the *HalluciGen* task to explore if and how models can detect and even generate hallucinated paraphrases and translations. To test (c) *consistency of output* in face of semantically equivalent but stylistically varied input, we defined the *Robustness* task where the similarity of the output for similar prompts is assessed. To test (d) *detection capacity of machine-generated* (as opposed to human-authored) text we defined the *Voight-Kampff* task where participating models were used to generate texts which were submitted to the PAN lab for classification as human-authored vs machine-generated.

These four quality characteristics are at the forefront of current discussions of *reliability* and *trustworthiness* of generative language models and the systems built to make use of such models. The four ELOQUENT tasks are all designed to be assessable using generative models, to be straightforward and simple to execute, and to require little human assessment effort.

ELOQUENT received 55 registration sign-ups for teams to participate in various subsets of the four tasks. Of these, 8 teams submitted experiment results. This is a high and frankly a disappointing attrition rate, and we will poll the participants to find out what might increase the likelihood of participants completing and submitting experiments. We have already identified: (1) Some potential participants who mostly work in research on deep learning or language technology are not aware of CLEF as an experimental venue (as opposed to e.g. Semeval or WMT) and we intend to move one lab to some other such venue as a bridging effort to bring in teams from adjacent fields to CLEF (2) Many potential participant teams are industrial development labs and have little experience in participating in shared tasks that they view as an academic exercise. This is at least partially addressed through the adoption of quality measures in commission-funded AI platform projects, where many industrial partners are keen to take part to have their offerings available on the platform. We will make concerted publicity efforts to raise awareness in both these above directions.

The goal of the ELOQUENT lab is to evaluate the quality of generative language models along four different axes: specificity of answers to in-domain questions, ability to detect and generate hallucinations, consistency of answers to linguistically varied input and reliability on distinguishing machine-generated from human-authored text. Overall, we find that the LLM quality is better according to some criteria (model hallucinations), rather than others (generating text that is indistinguishable from human-authored material). However, we also find that system performance varies highly for specific

task items, which makes systematic analysis quite challenging. We will be working in coming editions of ELOQUENT to combine human effort with self-assessment, and to systematically assess the reliability of the latter: we believe that such assessment will be a valuable outcome in its own right.

II. USAGE SCENARIOS AND DOMAIN

This type of evaluation finds its natural home at CLEF: Europe is a truly multilingual arena, with a large number of equally relevant languages and language varieties, dialects, and sociolects, none of which can be disregarded in commercial or public applications of information technology. A vibrant technology scene provides great readiness for deploying tools and services to a multi-lingual and increasingly mobile population with a high level of acceptance to information services. To provide inclusive services to a diverse population will require a large variety of background models with verified equivalent or comparable competence in several languages and language varieties.

Many practical application scenarios for generative AI require systems and models that are severely constrained by time or energy budget. This requires building parsimonious models using larger models as a guide which in turn requires a systematic verification of the quality of tailored, tuned, distilled, or extracted models compared to large all-round models.

The motivation for using a generative language model as a component in an information service is to handle a broad variety of input data robustly and elegantly, and to provide appropriately generated output to fit the varied application situations and the needs of a diverse user population. A classic gold standard-based evaluation may provide a useful sampling of such conditions, but validating this capability is better done in realistic situations which provide varied and unpredictable input as well as varied and unpredictable requirements for output.

Organisations, whether commercial or not, are unable to expend great human effort on testing, benchmarking, sampling, and validation to perform quality assurance on a language module which oftentimes is a subsidiary component of a larger envisioned service. Effective testing will therefore require automated or semi-automated testing procedures, and leveraging the capabilities of language models to self-assess its qualities.

III. ORGANISERS

This lab is co-organised by Silo AI, Toloka AI, MBZUAI, RISE Research Institutes of Sweden, the University of Crete, the University of Helsinki, the University of Oslo, Charles University, and the University of Edinburgh. Several of the organisers are engaged in various national, European, and international research and development projects, such as DeployAI, part of the AI on Demand Platform family of projects which brings together several industrial and academic research teams developing generative language models. This year, the consortium has broadened from the 2024 edition of ELOQUENT, with the addition of new tasks.

- **Ekaterina Artemova** is a senior research scientist at Toloka AI, holding a PhD from FRC CSC RAS. She co-organized the RuATD shared task on machine-generated text detection. Her research focuses on data-centric NLP, specifically on fair benchmarking strategies, low-resource settings, and LLM evaluation.
- **Ondřej Bojar** is an Associate Professor at ÚFAL, Charles University, and a lead scientist in Machine Translation in the Czech Republic. He has been co-organizing a well-known series of shared tasks in machine translation and machine translation evaluation (WMT) since 2013. Having taken part and later supervised ÚFAL's participation in a series of EU projects, he has recently concluded his coordination of the EU project ELITR (<http://elitr.eu/>) focussed on simultaneous speech translation into over 40 languages.
- **Stergios Chatzikyriakidis** is a Professor of Computational Linguistics at the University of Crete. His research interests lie at the intersection between NLP and Theoretical Linguistics. Some of the topics he has been working on include Natural Language Inference, Computational Dialectology, Constructive Type Theoretical Semantics and their Implementation, Probabilistic Semantics, Dialogue Modelling, and the interaction between Logic/symbolic methods and Machine Learning/Deep Learning for NLP.
- **Luise Dürlich** is a PhD student at Uppsala University, RISE and the Swedish Medical Products Agency with an interest in explainability, resource efficiency, and multilinguality.
- **Evangelia Gogoulou** is a PhD student at KTH in Stockholm and at RISE with a research interest in contextualised language models.
- **Liane Guillou** is a postdoctoral researcher in the StatMT group at The University of Edinburgh. The central themes of her research include evaluation, cross-linguality, and multilinguality. Liane was the lead organiser for the shared task on Cross-lingual Pronoun Prediction at the Conference on Machine Translation (WMT) in 2016, and a co-organiser for the pronoun test suite evaluation at WMT 2018 and the ELOQUENT lab at CLEF 2024. Most recently, she is currently involved in organising the Mu-SHROOM task on detecting hallucination and overgeneration at SemEval 2025.
- **Jussi Karlgren** has participated in every CLEF. He was a member of the CLEF Steering Committee from CLEF's foundation to 2009. He has been the co-organiser of iCLEF, the interactive search lab at CLEF (2006-2009), and the Podcast Track at TREC (2020-2021), and currently works on multi-lingual evaluation of generative language model quality in the DeployAI project.
- **Timothee Mickus** is a postdoctoral researcher at the University of Helsinki. His research centers around interpretability, distributional semantics and natural language generation, with a specific focus on generating definitions. He has been involved in several shared tasks, including CoDWoE (Comparing Dictionaries and Word Embeddings, Semeval-2021 Task 1), AXOLOTL'24 (As-

certain and explain overhauls of the lexicon over time at LChange'24) and SHROOM (Shared task on Hallucinations and Related Observable Overgeneration Mistakes, SemEval 2024 Task 6), and is currently the lead organizer of the Mu-SHROOM task at SemEval 2025

- **Vladislav Mikhailov** is a postdoctoral researcher in the Language Technology Group at University of Oslo. He was one of the lead organizers for the RuATD shared task. His main research interests include developing and benchmarking LLMs.
- **Preslav Nakov** is Professor at MBZUAI. Previously, he was a Principal Scientist at QCRI, where he led the Tanbih mega-project (in collaboration with MIT), which aimed to limit the effect of fake news, propaganda and media bias by making users aware of what they are reading, promoting media literacy and critical thinking. He has co-organized a total of 18 SemEval tasks, and other shared tasks at CLEF, WMT, ECIR, DiscoMT, NLP4IF, and VarDial. He co-chaired the SemEval Workshop in 2014-2016, and the shared tasks at IJCNLP-2017.
- **Joakim Nivre** is senior researcher at RISE and professor of computational linguistics at Uppsala University. He is one of the main developers of the transition-based approach to syntactic dependency parsing and a co-founder of the Universal Dependencies project, which develops annotated corpora for the world's languages and currently involves over 500 researchers around the world. He has extensive experience of organising shared tasks, in particular at the Conference on Computational Natural Language Learning (CoNLL), where he was responsible for shared tasks in 2007, 2008, 2009, 2017 and 2018. He is a fellow of the Association for Computational Linguistics and was the president of the association in 2017.
- **Lilja Øvrelid** is Professor at the Department of Informatics, University of Oslo and group leader for the Language Technology Group. She has extensive experience leading data annotation projects, such as the Norwegian Dependency Treebank (later UD treebank), the NARC coreference dataset and the NorQuAD Question-Answering dataset. She has contributed to the benchmarking of Norwegian LLMs through the NorBench initiative and was involved in organizing the 2017 and 2018 shared tasks on extrinsic parser evaluation and the SemEval 2022 shared task on structured sentiment analysis.
- **Magnus Sahlgren** is Head of research for NLU at AI Sweden. He has a PhD in computational linguistics, and has worked on computational models of meaning for more than 20 years in both research and industry. Sahlgren has participated in several CLEF campaigns, and has organised several international workshops.
- **Aarne Talman** is a language technology researcher at University of Helsinki, a Senior Manager of Data Science and Machine Learning at Accenture and the founder of Basement AI, an independent NLP R&D lab and consultancy. Aarne has close to 20 years of experience in the industry, including software engineering, consulting, and leadership. His research interests are focused on natural

language understanding, reasoning and large language models.

- **Erik Velldal** is a Professor at the Department of Informatics, University of Oslo. He has extensive experience leading data annotation projects, such as the NorNE dataset of named entities and the NoReC family of sentiment-related datasets. He has further contributed to the benchmarking of Norwegian LLMs through the NorBench initiative and was involved in organizing the 2017 shared task on extrinsic parser evaluation and the SemEval 2022 shared task on structured sentiment analysis.
- **Shorouq Zahra** is a research engineer at RISE in Stockholm.

IV. FORMAT AND SESSION

We plan to run the task with at least a three-year perspective. The first year has given us insights in procedures and methods, and we will develop the experimentation mechanisms accordingly. New ideas for tasks have been proposed to the organisers by many parties, both such that participated in the tasks and such that did not, and the general idea of organising shared tasks for generative language models to meet multilingual challenges and to address European values and the aspiration to trustworthiness and practical utility has resonated well in conversations with European model developers and with the European commission. We propose to hold the session at CLEF 2025 in a half day but would like to reserve the option to increase it if participation exceeds our expectations.

The session to be run in well-established CLEF and TREC style with a high-level overview of results and with most interesting (not necessarily the most highly scoring) participants invited to present their approaches in brief. We also plan that a fair slice of the time will be allocated to discuss the task structure and propose modifications and improvements to them, reacting both to technical, societal, and methodological developments.

V. AUDIENCE

The organisers have well established and extensive networks in academic research, industrial research, public policy, and commercial application of large language models and generative AI technology. Many CLEF participants are already working with large language models in the course of their experimentation with other evaluation labs and participating in this proposed task will require comparatively little marginal effort for them; the organisers have among them direct links to most research groups on generative language model development in Europe.

VI. TASKS

The tasks are planned to involve comparatively little human effort, and instead leverage the capacity of the current generation of AI models to generate, process, and assess human language. The first year’s discrepancy between registered participants and submissions indicated that in addition to a broader publicity effort we need a very simple task to draw in experimental work. For this purpose we will promote the a game-oriented task on *dictionary definition* which will require very little computation and we will keep the possibility of generating experimental data for the PAN lab as a continuation of last year’s Voight-Kampff task, which also has quite low participation thresholds.

The 2024 Robustness task will be evolved into a more multi-lingual format, and to take more systematic account of cultural factors such as politeness, formality, and personal tone.

For 2025, we introduce two new more demanding tasks: Preference Score Prediction and Reasoning. These tasks will involve more complex interaction with the system than the previous three tasks and the organisers will proceed by early involvement of teams that develop language models to ensure take-up.

- 1) **Dictionary Definition task** The dictionary game which is marketed in various languages under various names is based on a player picking a rare word from a dictionary and the other players writing sham definitions for the word. Each player then needs to identify which definition is the actual one and players gain points from fooling other players into believing their suggested definitions are the actual real one. We intend to formulate this as an entertaining entry level task for participant systems.
- 2) **Voight-Kampff Task** This task will ask participants in the ELOQUENT lab to generate material for the PAN lab task on distinguishing human authored texts from machine generated texts, a task which has considerably public, legal, and commercial interest. This task will, as the above Dictionary Definition task, be formulated and defined as a low threshold entry level task, which we will also promote to every team participating in other tasks.
- 3) **Robustness Task:** The task is to test the capability of LLMs to handle linguistic and cross-cultural variation as represented by human-generated varieties of input prompts. The results will be assessed by how variation in output is conditioned on variation of equivalent but non-identical input prompts. This task ran with almost exclusively English-language submissions in 2024 and we found some insights in the resulting data, but wish to investigate a broader and more systematic view of variation in this coming year.
- 4) **Preference Score Prediction Task:** Human preferences have become a standard paradigm for evaluating and aligning generative LLMs. However, the abilities of LLMs to reproduce the human feedback are still underexplored. In this task, we address the development of systems capable of predicting Likert scores across candidate model responses for a given prompt. We aim to create a novel dataset of human and model preferences spanning a wide range of evaluative criteria, including but not limited to fluency, relevance, factuality, and harmfulness. The human-based preference data will be collected from scratch, and the model-based preference data will be produced by popular open-source and proprietary LLMs.
- 5) **Reasoning task** Reasoning is one of the core capabilities expected of intelligent systems and large language models. The reasoning capability refers to their ability to process, analyze, and draw conclusions from complex information. Large language models are expected to perform various reasoning tasks, such as analogical reasoning, causal reasoning, and deductive reasoning. While current state of the art language models perform well in a number of reasoning tasks, as witness the performance in NLI tasks, reasoning in more complex situations, such as noisy dialogue environments, where a number of interjections, backchannels, split utterances are involved, is known to be difficult. In this task we test the reasoning capabilities of large language models by defining a set of tasks that tries to capture the wealth of reasoning patterns found in NL. We concentrate on a number of challenging issues: a) reasoning in noisy dialogue contexts, b) Enthymematic reasoning (reasoning under hidden premises), c) implicative reasoning, d) transitivity of inference, among others.
- 6) **Topical Test Generation Task:** This task ran in 2024 in the first edition of ELOQUENT. The idea of the task is for a given topic to generate a competence test in the form of a number of relevant questions. These questions are posed to the other models participating in the task, and the responses are then again scored by participating models. Quality of the test will be assessed by cross-model correlation of test scores. The feedback from participants from the first year of this task was that executing the experiment involved too many complex steps, and we will need to build a more streamlined process for handling submissions. The intention is to evolve this task into a test for *Conversational competence*, where two models or two instances of a model will be set up to engage in a conversation to solve some given task which will involve making use of both topical and conversational competence. We plan to defer this task to later potential editions of ELOQUENT pending refinement.
- 7) **HalluciGen Task:** The HalluciGen task on generating and detecting hallucinations, that ran successfully as part of ELOQUENT in 2024 and was initially intended to run for a second year, will be moved to different venue in 2025. Teams at ELOQUENT 2025 wishing to participate in the next iteration of HalluciGen will be redirected to the task once a venue has been confirmed.